QP CODE: 21100667

Reg No : .....

Name : .....

# B.Sc DEGREE (CBCS)EXAMINATION, MARCH 2021

## Third Semester

B.Sc Computer Science Model III

## Core Course - CC3CRT01 - DATABASE MANAGEMENT SYSTEMS

2017 Admission Onwards

68DA78DF

Time: 3 Hours Max. Marks : 80

### Part A

*Answer any ten questions.*

*Each question carries 2 marks.*

1. Define data model.
2. List out various DML commands.
3. Compare entity type and entity set.
4. Explain relationship attributes with an example.
5. Give the general form of SQL query.
6. Write the syntax of UPDATE query.
7. Write any four aggregate functions in SQL.
8. What is modification anomaly?
9. What are nested relations?
10. What are the different states of transaction?
11. Write any two functions of DBA.
12. What is the use of GRANT command in SQL?

(10×2=20)

### Part B

*Answer any six questions.*

*Each question carries 5 marks.*

13. What are the different types of database end users? Discuss the main activities of each.

14. Draw component modules of a DBMS and their interactions.

15. What are the main characteristics of relations?

16. Discuss the characteristics of relations that make them different from ordinary tables and files.

17. Explain SET operations available in SQL.

18. Briefly describe nested queries in SQL.

19. Explain 2NF and 3NF with suitable examples.

20. What is the difference between dense index and sparse index?

21. Explain for main control measures to provide security of data in databases.

(6×5=30)

## Part C

*Answer any two questions.*

*Each question carries 15 marks.*

22. Explain in detail about three schema architecture and data independence.

23. Construct an E-R schema diagram for the company database . Details of COMPANY database is as follows. (1) The company is organised into departments.each department has a unique name,unique numberand a particular employee who manages the department.Keep track of the start date when that employee began managing the department.A department may have several locations. (2) A department controls a number of projects,each of which has a unique name,a unique number and a single location. (3) Employee has name,SSN,address,salary,gender and birth date.An employee is assigned to ine department but may work on several projects,which are not necessarilyb controlled by the same department.Keep track of the current number of hours per week that an employee works on each project.Also keep track of the direct supervisor of each employee. (4) Keep track of the dependants of each employee for instance purposes.Dependant has name,genderdob and relationship to employee.

24. Explain various types of Normal Forms with examples.

25. Explain discretionary access control based on granting and revoking previleges.

(2×15=30)